Dynamic programming exercises

15-E01 Rod-cutting revenue

Difficulty: Basic

p=[0,2,5,7,10,12] gives prices for lengths1..5. Compute r1..r5 and an optimal cutting of length5.

15-E02 Matrix-chain transition

Difficulty: Advanced

Matrices have dimensions2×3,3×4,4×2. Compare scalar multiplication counts for both full parenthesizations and confirm the DP optimum.

15-E03 LCS transitions and reconstruction

Difficulty: Basic

X=ABA,Y=BAA. Give the full LCS length table, optimal length, and one reconstructed LCS.

15-E04 Global alignment states and reconstruction

Difficulty: Research

X=ACGTA,Y=AGCAC. Match earns3,mismatch−2,and each insertion/deletion gap−3. Establish optimal substructure, give the full score table, reconstruct an optimum, and explain why equal-length prefixes alone are insufficient.

15-E05 Complete unbounded-knapsack DP

Difficulty: Research

Items A=(weight3,value5),B=(4,7),C=(7,12) have unlimited supply. With total weight at most13, find the exact optimum and items. Check highest value/weight greedy.

15-E06 Optimal BST with unequal failure probabilities

Difficulty: Advanced

Sorted keys k1<k2 have p1=.10,p2=.20 and failure probabilities q0=.50,q1=.10,q2=.10. Count dummy checks. Give e,w,root, reconstruct the tree, and verify expected cost by depths. Must the highest-success-probability key be root?

Dynamic programming exercises (continued)

15-E07 Failure of endpoint-only longest-simple-path states

Difficulty: Research

Directed unit-weight edges are u→z,z→v,v→z,z→w. L(a,b) stores only endpoints for longest simple paths. Find L(u,v),L(v,w), test their concatenation through v, and give a DP state preventing repeated vertices.

> Complete your derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

Lin Cing
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